



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Production Engineering
Summer Term Examination
Academic Year: 2024-25

Class: Second Year	Subject: Mechanical Working of Metals	Subject Code: PCC-PE301
Date: 18/06/2025	Time: 10:00 -13:00	Max. Marks : 100

Note

- i. Attempt all questions.
- ii. Assume suitable data whenever necessary and state it clearly.
- iii. Figures to right indicate full marks.
- iv. The symbols carry usual meaning.

Q.1	Attempt any FOUR of the following	M	CO	BT
a	Summarize the merits and demerits of hot working and cold working.	05	CO1	BT2
b	Compare engineering strain and true strain.	05	CO1	BT2
c	Define engineering stress and true stress.	05	CO1	BT1
d	Explain theory of Plasticity.	05	CO1	BT2
e	Compare the differences between hot working and cold working processes.	05	CO1	BT2
Q.2	Attempt any FOUR of the following.			
a	Summarize the economic considerations in sheet metal working operations	05	CO2	BT2
b	Classify the sheet metal working operation and clearly indicate the stresses induced.	05	CO2	BT2
c	Explain the mechanism of shearing in sheet metal operation with neat sketch.	05	CO2	BT2
d	Discuss briefly about the defects in sheet metal products.	05	CO3	BT2
e	Define drawing and explain cup and tube drawing process with proper sketches.	05	CO2	BT2
Q.3	Attempt any FOUR of the following.			
a	Explain the concept of spring back in sheet metal bending with a suitable sketch.	05	CO2	BT2
b	Outline the working principle of coining and ironing process with neat sketch.	05	CO3	BT2
c	Explain the injection moulding process in detail with suitable sketch.	05	CO3	BT2
d	Explain the working principle of thermoforming process with neat sketch.	05	CO3	BT2
e	Summarize the characteristics of drawing process.	05	CO3	BT2
Q.4	Attempt any FOUR of the following.			
a	Explain the principle and theory of rolling process with a schematic diagram.	05	CO4	BT2
b	Define forging. Explain smith forging processes with the necessary sketches.	05	CO4	BT1

c	Outline the working principle of ring rolling process with a neat sketch.	05	CO3	BT2
d	Discuss in detail about the compression moulding process with a neat sketch.	05	CO4	BT2
e	Write down the classification of extrusion processes.	05	CO3	BT2
Q.5	Solve the following questions.			
a	Classify the types of rolling mills with the necessary sketches.	10	CO3	BT6
b	Discuss in detail about the various types of forging hammers.	10	CO4	BT2

=== ALL THE BEST ===

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Classify different metal forming processes.
CO 2	Design dies for cutting, forging, bending, drawing, piercing, and blanking.
CO 3	Compute material properties, forces and defects in metal forming processes
CO 4	Select equipment for metal forming processes.

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Production Engineering
Summer Term Examination
Academic Year: 2024-25

Class: Second Year	Subject: Mechanical Measurement and Metrology	Subject Code: PCC-PE208
Date: 25/06/2025	Time: 14:30 -17:30	Max. Marks : 100

Note

- Attempt all questions.
- Assume suitable data whenever necessary and state it clearly.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.1	Attempt any FOUR of the following	M	CO	BT
a	What is measurement? Give its types.	05	CO1	BT1
b	Distinguish between line standard and end standard.	05	CO1	BT2
c	Describe the different types of errors in measurement and their causes.	05	CO1	BT2
d	What is the need of calibration? Explain	05	CO1	BT1
e	With suitable example, explain the difference between precision and accuracy.	05	CO1	BT2
Q.2	Attempt any FOUR of the following.			
a	Define a comparator. Discuss the functional requirements of a comparator.	05	CO3	BT1
b	Classify the comparator according to the principles used for obtaining magnification.	05	CO3	BT1
c	Explain the working mechanism of a dial indicator.	05	CO3	BT2
d	Explain with the neat sketch working principle of a solex pneumatic gauge.	05	CO3	BT2
e	Explain with the neat sketch working principle of a sigma mechanical comparator.	05	CO3	BT2
Q.3	Attempt any FOUR of the following.			
a	Explain with neat sketch optical bevel protractor.	05	CO4	BT2
b	What is mean by wringing process? Describe briefly grades of slip gauges.	05	CO4	BT1
c	Explain with neat sketch clinometer.	05	CO4	BT2
d	Explain with neat sketch universal bevel protractor.	05	CO4	BT2
e	Determine the elevation for 42° using 5.00" sine bar.	05	CO4	BT2
Q.4	Attempt any FOUR of the following.			
a	Name the different terms used in surface roughness.	05	CO5	BT1
b	Briefly describe the construction, principle, and operation of Talysurf with a neat sketch.	05	CO5	BT1
c	Define manometer? Elaborate the U- tube Manometer in briefly.	05	CO6	BT2

d	Explain about Diaphragm gauge in detail. write advantages.	05	CO6	BT2
e	Explain the working principle Bourdon tube pressure gauge.	05	CO6	BT2
Q.5	Solve the following questions.			
a	Construct in detail the working of the Sine Bar to measure unknown angle.	10	CO4	BT6
b	Express the following methods of qualifying surface roughness: (a) Ra value. (b) RMS value. (c) Rz value	10	CO5	BT2

=== ALL THE BEST ===

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Use linear and angular measuring instruments.
CO 2	Design tolerances and fits for selected product quality.
CO 3	Identify suitable comparator and light waves for different measurements.
CO 4	Understanding of basic concepts of mechanical measurement and errors in measurements.
CO 5	Measure various machining process by measuring surface finish of the component produced workout for surface integrity parameters.
CO 6	Select appropriate temperature and pressure measuring device for various applications.

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